Table 8: A summary of the fat soluble vitamins

Vitamin	Functions	RDA	Deficiency Symptoms	Toxicity
Vitamin A	Aids in vision, especially night vision promotes growth, development, reproduction, healthy skin cells, and health immune system acts as an antioxidant	Men : 1000 retinol equivalents (RE) 5000 1μ	Short – term: Night blindness, flaking skin. Long term : Xerophthalmia (dry, hard cornea); progresses to softening of corneas and blindness if untreated	Toxic symptoms appear with chronic intake of 10 times the RDA, usually due to supplement abused. Symptoms include lack of appetite; dry, itch skin; loss and coarsening of hair; liver damage; fetal defects and miscarriage.
Vitamin D	Aids in absorption, transport, and use of calcium and phosphorus Maintains optimal blood calcium levels	5 – 10 μg	Rickets (faulty bone formation) in children. Osteromalacia (Softening of bones) in adults	Toxic symptoms appear with regular intake of 5 times the RDA, usually due to supplement abuse. Symptoms include lack of appetite, high blood pressure, excessive thirst and urination, high blood calcium and calcium deposits in tissues, kidney stones, and kidney failure can lead to death.
Vitamin E	Acts as an antioxidant, protecting cell membranes from oxidative damage.	20 – 25 mg	Damage to all membranes, leading to destruction of red blood cells (hemolyticanemi) Rare in humans except in premature infants and those with certain diseases	Relatively nontoxic; use of supplements with 80 times or more of the RDA may cause symptoms. Symptoms include muscular weakness, fatigue, and nausea.
Vitamin K	Promotes formation of some blood – clotting factors	Men : 70 – 80 μg Women : 60 – 65 μg	Poor blood clotting, possibly leading to hemorrhage (Rare in humans excepts in newborn infants and people on long term general antibiotic therapy)	Natural forms are nontoxic; excessive synthetic supplementation can cause toxic symptoms in infants. Symptoms include anemia and jaundice.

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